



Mechatronics Research Group

CENTRE FOR ADVANCED STUDIES

Dr. A.P.J. Abdul Kalam Technical University Lucknow

Website: www.cas.res.in



Research Seminar Series

Seminar -12

Corrosion Prevention of Steel using Arbi Leaves

DATE & TIME:

**22th January 2022,
12:00 PM**

PLATFORM & WEB-LINK

Google Meet

(<https://meet.google.com/adx-djsg-cff>)



SPEAKER

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ABSTRACT

Every year, corrosion of metals causes a great loss to the economy and puts many lives in danger. These losses can be avoided by a variety of ways; however, use of a biodegradable inhibitor can be a better option for this purpose. The present work reports a similar kind of biodegradable inhibitor (*Colocasia esculenta* leaves) for low carbon steel (LCS) prevention in 0.5 M NaCl. *Colocasia esculenta* leaves aqueous extract (AECEL) is examined by UV-visible spectroscopy (UVS) and FTIR spectroscopy (FTIRS), weight loss measurements (WL), open circuit potential (OCP) curves, Tafel polarization curves (TPC), electrochemical impedance spectroscopy (EIS) and scanning electron microscopy (SEM). The analysis of AECEL by UVS and FTIRS disclose that AECEL possesses several biomolecules. These biomolecules have the ability to be adsorbed on LCS and thus can provide immunity to LCS against corrosion. Based on examination the maximum inhibition achieved is 76% (TPC). The